
TCX2005: Information Systems, Management and Organisations

L6: Critical IS Infrastructure-I

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Recap

Information Systems Infrastructure

Organizational Needs

Reasons for adopting IS

Infrastructure Components

The core elements enabling IS

Implementation Challenges

Hurdles in deploying IS

Competitive Advantage

Benefits gained from IS



Lesson Objectives



Core infrastructure components



Trends in hardware & software
platforms



Key challenges in managing IT/IS
infrastructure

Learning Resources



- Textbook (read before class)
 - Read chapter 5
- Any Other Material
 - Shared on Canvas for this week

Case Study

Read the case study from textbook on –[Southeast Asia's Grab Leverages IT to Enhance Its Services](#):

Identify:

- a. The problem
- b. The solution
- c. Outcomes
- d. Key learning point(s)

Note: be ready to share your thoughts

Case Study

Problem:

- Unreliable and unpredictable taxi services
- Lack of transparent pricing (especially during peak periods)
- Limited payment options in a region with low bank account penetration

Solution:

Comprehensive IT/IS infrastructure

- Cloud Infrastructure: Amazon Redshift and ElastiCache for real-time data management
 - Analytics Platform: Geohashing technology for spatial analysis
 - Mobile Applications: Integrated Android and iOS apps with real-time updates
 - Payment Systems: E-payment platform partnership with Lipso
 - Unified technology infrastructure across 339 cities
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Case Study

Outcomes:

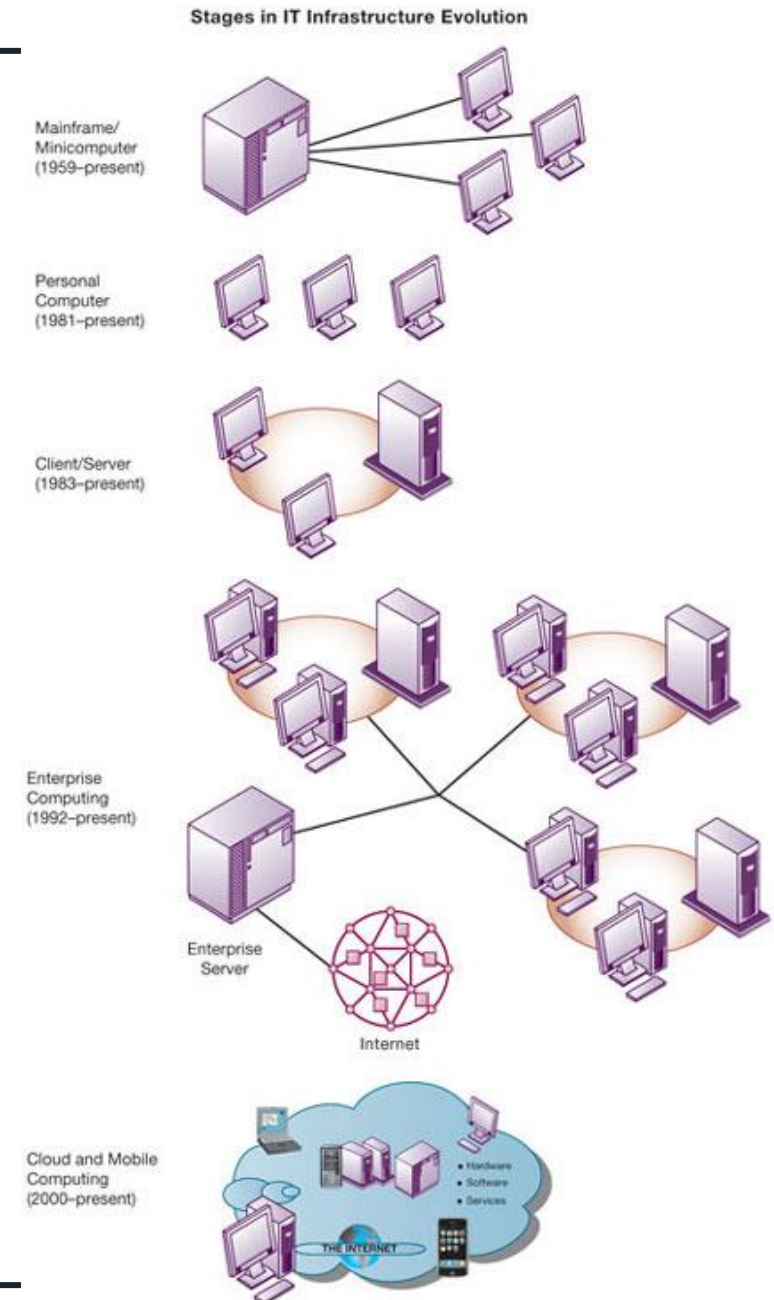
- Operational Efficiency managing nearly 6 million daily rides and 2.8 million drivers
- Enhanced service reliability
- Improved resource allocation
- Expanded financial inclusion
- Regional market leadership

Learning:

- IT/IS directly enhance organizational performance by supporting essential business processes and generate value for multiple stakeholders (customers, drivers, merchants etc.).
 - Effective IT/IS infrastructure allows organizations to manage finite resources dynamically.
 - Modern infrastructure enables real-time data processing and decision-making
 - Infrastructure decisions have long-term strategic implications.
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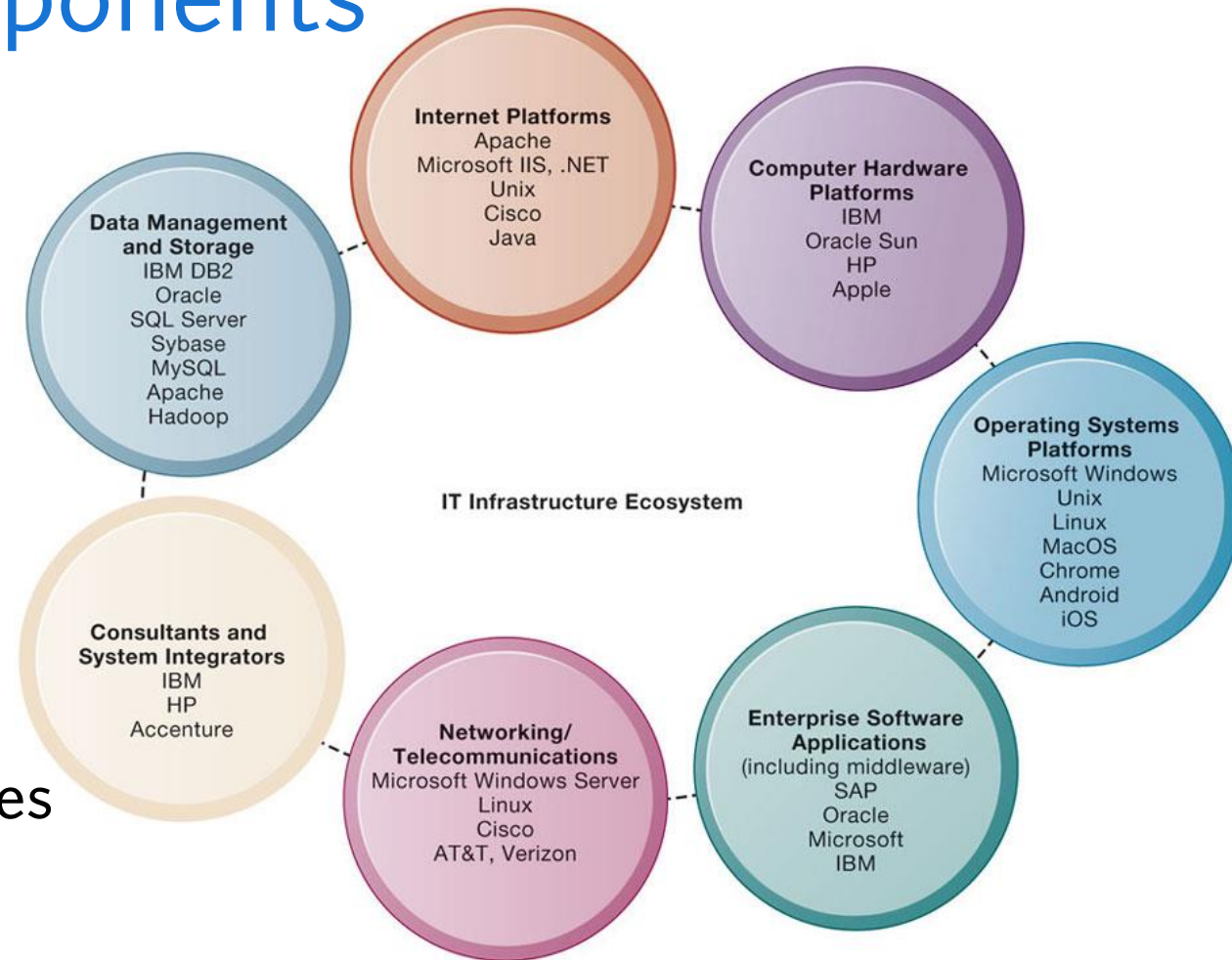
Evolution of IT Infrastructure

- Stage 1: Mainframe Era (1959- onward)
 - Centralized computing
 - Terminal-based access
 - Example: DBS Bank adopted IBM's z15 mainframe technology to enhance their digital transformation efforts
- Stage 2: Personal Computer Era (1981-present)
 - Decentralized computing
 - Individual productivity tools
 - Example: NUS' first PC labs in 1980s
- Stage 3: Client/Server Era (1980s-present)
 - Networked computing
 - Shared resources
 - Example: Singapore's first LAN deployments
- Stage 4: Enterprise Computing (1990s-present)
 - Internet-enabled business
 - Enterprise applications
- Stage 5: Cloud & Mobile Computing (Mid 2000s-present)
 - Cloud-based services
 - Mobile-first approach
 - Example: GovTech's SGTS (Singapore Government Tech Stack)



Core Infrastructure Components

1. Data management and storage
2. Internet platforms
3. Computer hardware platforms
4. Operating system platforms
5. Enterprise software applications
6. Networking/telecommunications platforms
7. Consulting system integration services



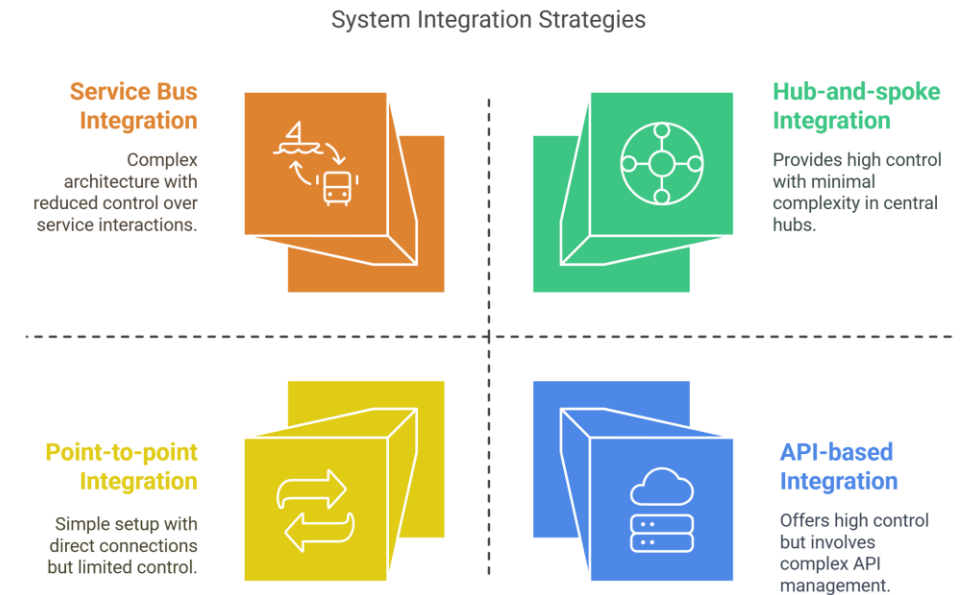
System Integration Services

Integration types:

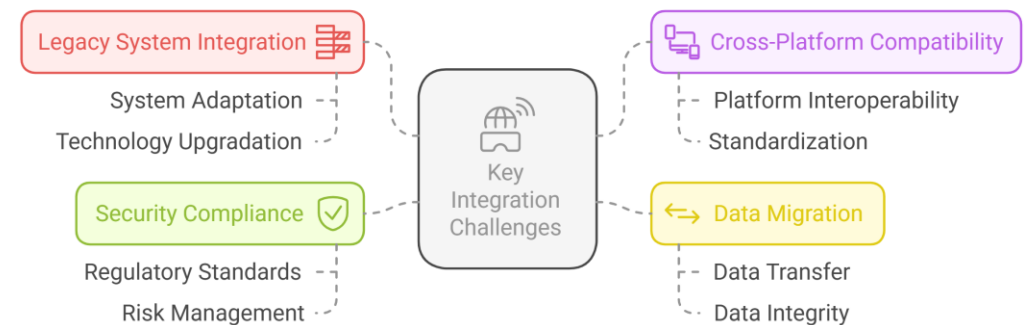
- Point-to-point
- Hub-and-spoke
- Service bus
- API-based

Challenges

- Legacy system integration
- Cross-platform compatibility
- Data migration
- Security compliance



Addressing Key Integration Challenges

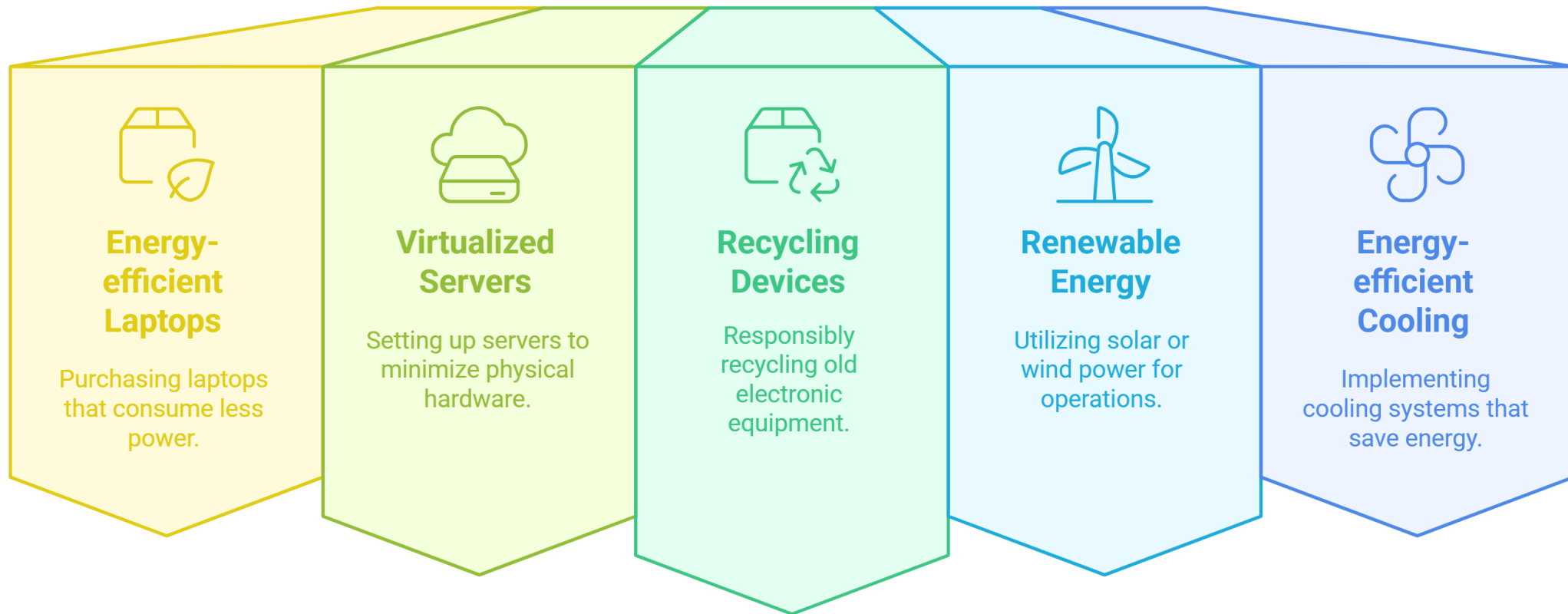


Current Trends in Computer Hardware Platforms

- Cloud computing
 - On-demand computing services obtained over network → reduced costs
 - E.g., IaaS, SaaS, PaaS
 - Edge computing
 - Servers at the edge of the Internet (located closer to data sources)
 - Reducing latency, and network traffic
 - Green computing (Green IT)
 - Practices and technologies for manufacturing, using, disposing of computing and networking hardware
 - High performance, power-saving processors
 - Power-efficient multicore processors
 - The mobile digital platform
 - Smartphones, Netbooks, wearables etc.
 - Consumerization of IT and BYOD (bring your own device)
 - Forces businesses and IT departments to rethink how IT equipment and services are acquired and managed
 - Quantum computing
 - Uses quantum physics to represent and operate on data
 - Dramatic increases in computing speed
 - E.g., “Majorana 1” chip by Microsoft (February 2025)
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Current Trends in Computer Hardware Platforms

Green Computing Practices

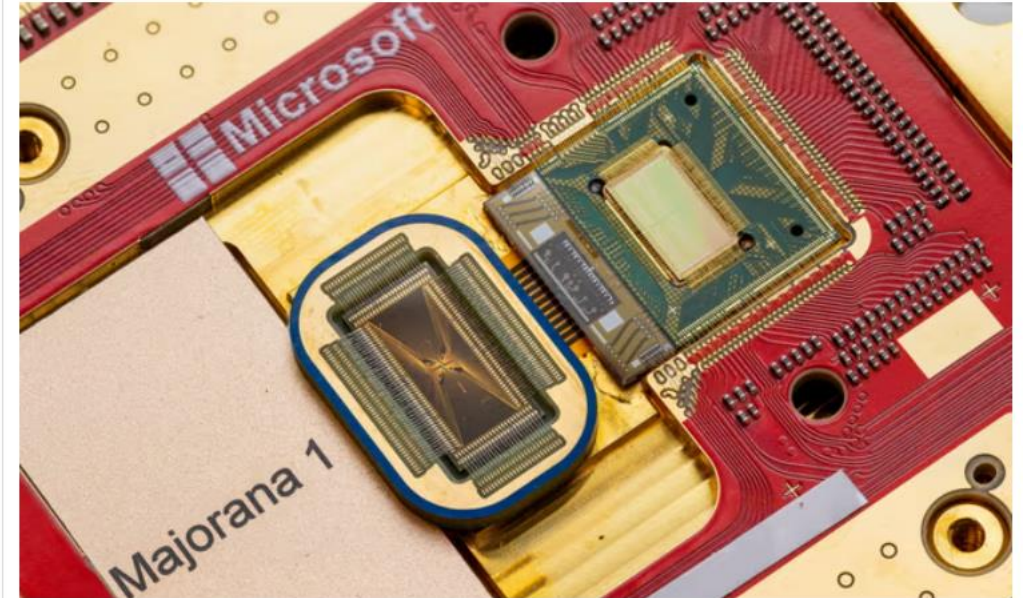


Discussion

- If quantum computing becomes mainstream, how could it impact businesses, governments, and everyday technology users?
- If quantum computing becomes widely accessible in the next decade, what industries will be most disrupted, and what new challenges might arise?

Microsoft unveils chip it says could bring quantum computing within years

Chip is powered by world's first topoconductor, which can create new state of matter that is not solid, liquid or gas



Microsoft claims that the topoconductor offers a path to developing quantum systems that can scale to a million qubits. Photograph: John Brecher

[source](#)

Current Trends in Computer Software Platforms

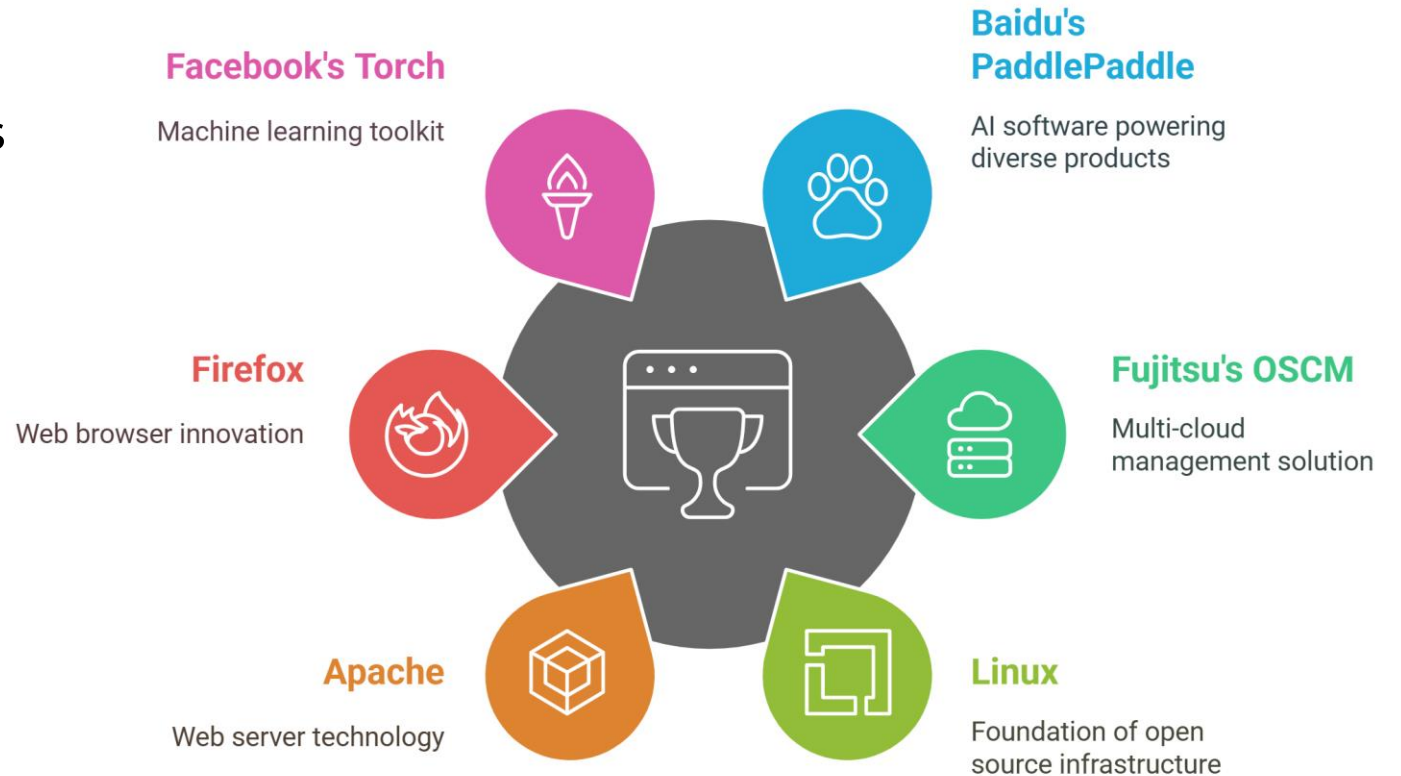
- Software outsourcing and cloud services
 - Software packages and enterprise software
 - Software outsourcing
 - Cloud-based software services and tools
 - Distributed computing
 - Blockchain platforms
 - Others:
 - Artificial Intelligence/ML
 - Zero-trust security
 - Container orchestration
- Linux and open-source software
 - Produced by community of programmers
 - Examples: Apache web server, Mozilla Firefox browser, OpenOffice
 - Software for the web
 - Java Virtual Machine
 - Web browsers
 - HTML and HTML5
 - Web services and service-oriented architecture
 - Web services
 - XML : Extensible Markup Language
 - SOA : service-oriented architecture
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Discussion: Free Lunch??

On the right you see example of open sourced resources by various companies

- How do companies benefit from releasing their proprietary software as open source?

Open Source Contributions to Competitive Advantage



Key Challenges in Modern Enterprise IT/IS

1. Platform and Infrastructure Change

- As firms shrink or grow, IT needs to be flexible and scalable
- Scalability
 - Ability to expand to serve larger number of users
- For mobile computing and cloud computing
 - New policies and procedures for managing these new platforms
 - Contractual agreements with firms running clouds and distributing software required

2. Governance

- Who controls IT infrastructure?
 - How should IT department be organized?
 - Centralized
 - Central IT department makes decisions
 - Decentralized
 - Business unit IT departments make own decisions
 - How are costs allocated between divisions, departments?
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Key Challenges in Modern Enterprise IT/IS

3. Infrastructure Investment

- Under-investment and over-investment can hamper firm performance
 - Investment Considerations:
 - Rent vs. Buy decisions
 - Cloud vs. On-premise solutions
 - Security requirements
 - Impact on business processes and workflow
 - Ability to focus on core businesses rather than technology issues
 - Balancing cost with performance needs
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Total Cost of Ownership (TCO) Model

- Analyzes direct and indirect costs
- TCO can be reduced
 - Use of cloud services, greater centralization and standardization of hardware and software resources

Direct Costs

- Hardware acquisition
- Software licenses
- Installation
- Training

Indirect Costs

- Support and maintenance
- Infrastructure management
- Downtime costs
- Space and energy

Hardware, software account for only about 20% of TCO!!; Consider full TCO in investment decisions

Class Activity

TCO analysis
